ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Mathematics practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 B ✓ 1.6 D ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ [10]

QUESTION 2

2.1 (a) Yes, it ends in 5 ✓ (b) No, the sum of its digits (2 + 3 + 4 + 5 = 14) is not divisible by 3 ✓ (2)

2.2 $(56 + 44) \times 25$ (distributive property) ✓ $= 100 \times 25 = 2\,500$ ✓ (2)

2.3 (a) 24×36 ✓ $= 864$ chairs ✓ (b) $1\,000 - 864 = 136$ people ✓ (3)

2.4 (a) $30\% \times \text{R}12\,000 = \text{R}3\,600$ ✓ (4)

(b) $^{1}/_{8} \times \text{R}12\,000 = \text{R}1\,500$ ✓

(c) $\text{R}12\,000 - (\text{R}3\,600 + \text{R}3\,600 + \text{R}1\,500)$ ✓ $= \text{R}12\,000 - \text{R}8\,700 = \text{R}3\,300$ ✓

2.5 $\text{Interest} = \text{R}5\,000 \times {}^{12}/_{100} \times 2$ ✓ $= \text{R}1\,200$ ✓ $\text{Total} = \text{R}6\,200$ ✓ (3)

[14]

QUESTION 3

3.1 2^4 ✓ $\times 3^2$ ✓ (2)

3.2 (a) $25 - 8$ ✓ $= 17$ ✓ (b) $\sqrt{100}$ ✓ $= 10$ ✓ (c) 7×8 ✓ $= 56$ ✓ (6)

[8]

QUESTION 4

4.1 (a) $^{7}/_{8} - {}^{2}/_{8} + {}^{4}/_{8}$ ✓ $= {}^{9}/_{8} = 1\,{}^{1}/_{8}$ ✓ (4)

(b) $^{4}/_{3} \times {}^{3}/_{8}$ ✓ $= {}^{12}/_{24} = {}^{1}/_{2}$ ✓

4.2 (a) 2 070 ✓ (b) $56 \times 7 = 392$ ✓ $\rightarrow 0{,}392$ ✓ (3)

4.3 (a) $^{35}/_{100} = 0{,}35$ ✓ (b) 35% ✓ (2)

4.4 20% of $\text{R}250 = \text{R}50$ ✓ New fare $= \text{R}300$ ✓ (2)

4.5 $^{3}/_{5} = 60\%$ ✓; $68\% > 60\%$, so the second test ✓ (2)

4.6 $^{2}/_{3}$ (0,667) ; 65% ; 0,606 ; 0,6 ✓ (1)

[14]

QUESTION 5

5.1 1 ✓ ; 5 ✓ (2)

5.2 (a) Sutherland ✓ (b) $3 - (-7)$ ✓ $= 10$ ℃ ✓ (3)

5.3 $-9 + 4 + 2 = -3$ ✓ (1)

[6]

QUESTION 6

6.1.1 (a) 17 ✓ (b) 41 ✓ (2)

6.1.2 Number of tiles $= 4n + 1$ ✓✓ (2)

6.1.3 $4n + 1 = 81 \rightarrow 4n = 80$ ✓ $\rightarrow$ Pattern 20 ✓ (2)

6.2 (a) $5 \times 3 - 1 = 14$ ✓; $7 \times 3 - 1 = 20$ ✓ (4)
(b) $29 + 1 = 30$ ✓; $30 \div 3 = 10$ ✓

6.3 (a) $x = 18$ ✓ (b) $2x = 18$ ✓ $\rightarrow x = 9$ ✓ (3)

6.4 (a) $m + 4$ ✓ (3)
(b) $m + m + 4 = 30$ $(2m + 4 = 30)$ ✓ $\rightarrow m = 13$ years ✓

[16]

QUESTION 7

7.1 Open the compass to more than half the length of AB and draw arcs above and below AB from A ✓. With the same radius, draw arcs from B to cut the first arcs ✓. Join the two points where the arcs cross; this line bisects AB (and is perpendicular to it) ✓. (3)

7.2 $\angle B = (180° - 90°) \div 2$ ✓ $= 45°$ ✓ (2)

7.3 Similar ✓; the sides are in the same ratio (each side of ΔXYZ is double the matching side of ΔPQR) but the sizes are different ✓ (2)

7.4 (a) Rectangle ✓ (b) Square ✓ (2)

7.5 90° ✓ clockwise ✓ (2)

7.6 8 faces (2 hexagons and 6 rectangles) ✓ (1)

[12]

QUESTION 8

8.1.1 $P = 2(15 + 8)$ ✓ $= 46$ m ✓ (2)

8.1.2 $46 \times R85$ ✓ $= R3\ 910$ ✓ (2)

8.1.3 $15 \times 8 = 120$ m^2 ✓ (1)

8.1.4 $120 \div 25 = 4{,}8$ ✓ $\rightarrow 5$ bags (round up) ✓ (2)

8.2 $24 = ½ \times 8 \times h \rightarrow 24 = 4h$ ✓ $\rightarrow h = 6$ cm ✓ (2)

8.3 $V = 2 \times 1{,}5 \times 1$ ✓ $= 3\ m^3$ ✓ $= 3\ 000\ \ell$ ✓ (3)

[12]

QUESTION 9

9.1.1 5 learners ✓ (1)

9.1.2 56 ✓ (1)

9.1.3 53 (the 7th value) ✓ (1)

9.1.4 $69 - 35 = 34$ ✓ (1)

9.2.1 $^3/_8$ ✓ (1)

9.2.2 $^6/_8$ ✓ $= {}^3/_4$ ✓ (2)

9.2.3 $^3/_8 \times 80 = 30$ times ✓ (1)

[8]

TOTAL: 100

END OF MARKING GUIDELINE